

# Chapter 102

# Flow Organizational Nodes

This chapter details the Groups, Macro, and Pipe Router nodes, which are designed to help organize your compositions, making the node tree easier to see and understand.

For purposes of this document, node trees showing MediaIn nodes in DaVinci Resolve are interchangeable with Loader nodes in Fusion Studio, unless otherwise noted.

## Contents

|                           |      |
|---------------------------|------|
| <b>Groups</b> .....       | 2290 |
| Groups Introduction.....  | 2290 |
| Usage.....                | 2290 |
| <b>Macro</b> .....        | 2291 |
| Macro Introduction.....   | 2291 |
| Usage.....                | 2291 |
| Macro Editor .....        | 2291 |
| The Final Macro .....     | 2292 |
| <b>Pipe Router</b> .....  | 2292 |
| Router Introduction ..... | 2292 |
| Usage.....                | 2293 |
| Router .....              | 2293 |

# Groups



The Group node

## Groups Introduction

Groups are used to keep complex node trees organized. You can select any number of nodes in the node tree and then group them to create a single node icon in the Node Editor. Groups are non-destructive and can be opened at any time.

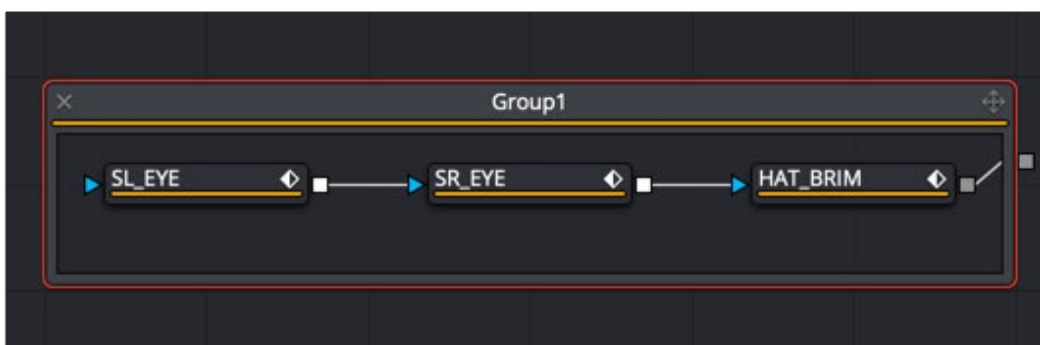
## Usage

- To group nodes, select them in the Node Editor, and then right-click over any of the selected nodes and choose Group from the contextual menu.



Selecting nodes to group

- To edit the individual nodes in a group, right-click and choose Expand Group from the contextual menu. All individual nodes contained in the group are displayed in a floating node tree window. When opened, groups hover over existing elements, allowing editing of the enclosed nodes.



An opened node group

- To remove or decompose a group and retain the individual nodes, right-click the group and choose Ungroup.

# Macro

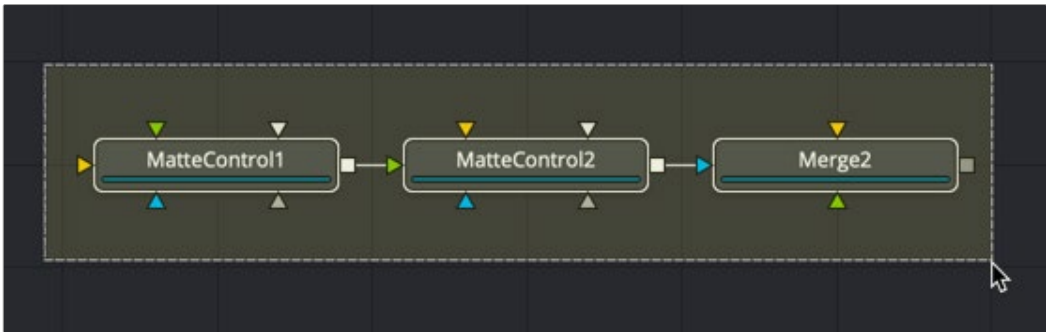
Macros are not technically a node. Instead, they are a group of nodes that act as a single node.

## Macro Introduction

Macros can be used to combine multiple nodes and expose a user-definable set of controls. They are meant as a fast and convenient way of building custom nodes.

## Usage

To create a Macro, select the nodes intended for the macro. The order in which the nodes are selected becomes the order in which they are displayed in the Macro Editor. Right-click on any of the selected nodes and choose Macro > Create Macro from the contextual menu.

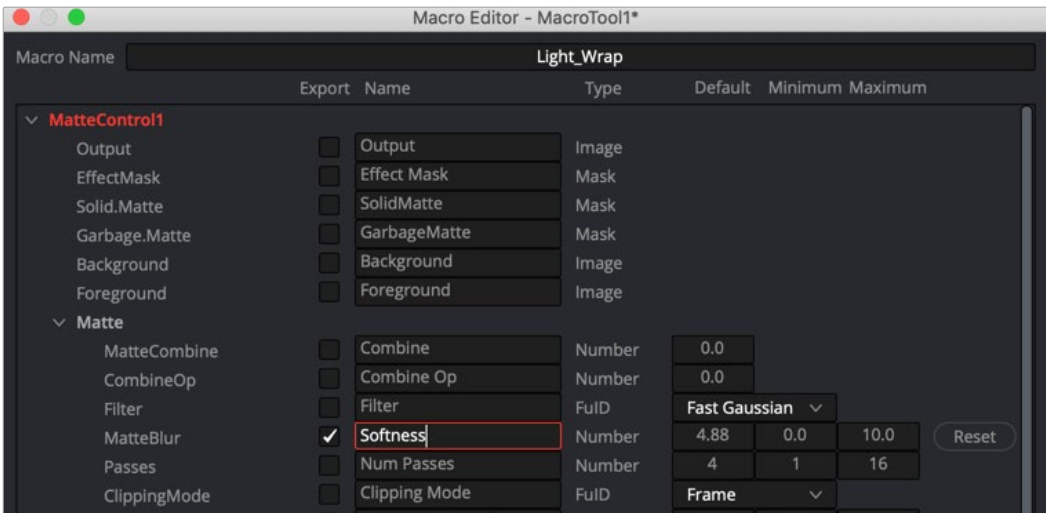


Select the nodes to include in the macro

## Macro Editor

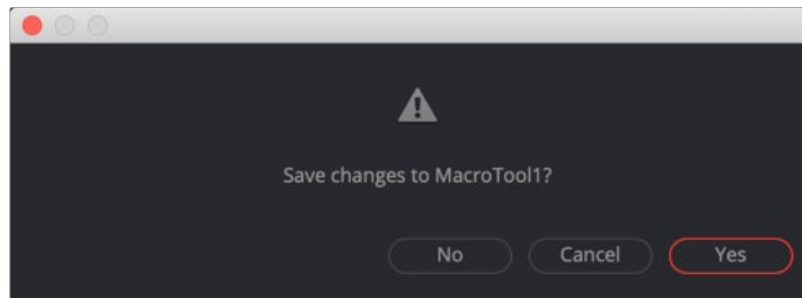
The Macro Editor allows you to specify and rename the controls that are exposed in the final macro tool.

In the example below, the tool is named Light\_Wrap at the top. The Blur slider for Matte Control 1 is enabled and renamed to Softness, as it will appear in the Inspector.



The Macro Editor

After setting up the macro to your liking, click the Close button in the lower-right corner of the dialog. Then, in the Save dialog, click Yes to save the macro, click No to leave Macro Editor without saving the changes, or click Cancel to return to the Macro Editor.



The Macro Editor save dialog

To add the macro to your node tree, right-click anywhere on the node tree and select Macro > [NameOfYourMacro] from the contextual menu.

## Saving a Macro as a Title Template in the Edit Page (DaVinci Resolve)

When using DaVinci Resolve, macros are available only in the Fusion page. However, if the macro is a title animation, you can save it to the Titles Templates folder and have it appear in the Edit page Effects Library.

To save a title macro so it appears in the Edit page Effects Library, save the macro to:

**macOS:** Users > UserName > Library > Application Support > Blackmagic Design > DaVinci Resolve > Fusion > Templates > Edit > Titles

**Windows:** C Drive > Users > UserName > AppData > Roaming > Blackmagic Design > DaVinci Resolve > Support > Fusion > Templates > Edit > Titles

## The Final Macro

The final macro looks and behaves just like any other node in Fusion.

As another example, you could take a single Channel Boolean, set it to Add mode, and make it into a macro exposing no controls at all, thus creating the equivalent of an Add Mix node like the one that can be found in programs like Nuke.

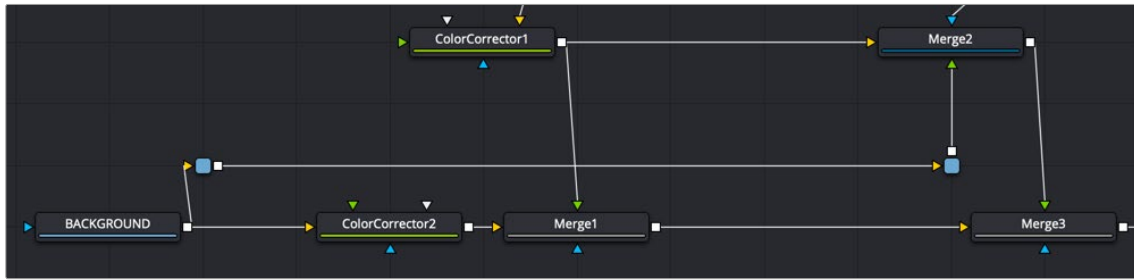
# Pipe Router

Pipe Routers are another type of organizational tool you use to improve the layout and appearance of the node tree.

## Router Introduction

Routers can be used to neatly organize your comps by creating “elbows” in your node tree, so the connection lines do not overlap nodes, making them easier to understand. Routers do not have any influence on render times.

## Usage

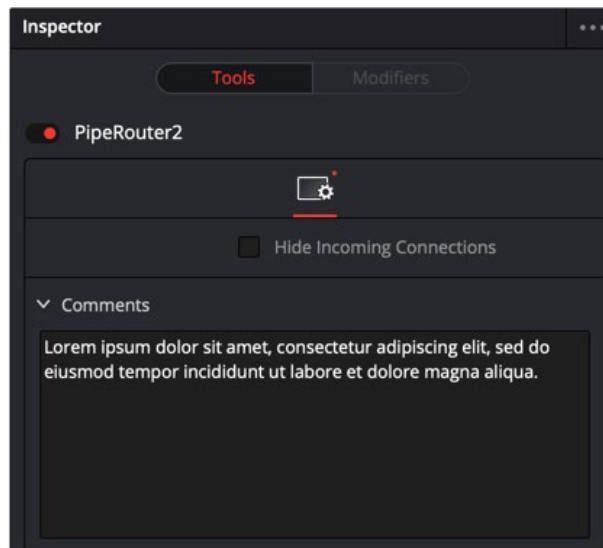


A Pipe Router usage example

## Router

To insert a router along a connection line, Option- or Alt-click on the line. The router can then be repositioned to arrange the connections as needed.

Although routers have no actual controls, they still can be used to add comments to a comp.



An example comment in a Router node